

The Digital Aquaculture *Nutrition Book*

Modern methods of raising and culturing fish and shrimp — and the modern nutrition concepts that power them: from larval live feeds to RAS, biofloc, functional feeds and precision feeding.

FISH NUTRITION

SHRIMP NUTRITION

FEED MANUFACTURING

RAS · BIOFLOC

PRECISION FEEDING

SUSTAINABILITY

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THE DIGITAL NUTRITION BOOKS

The Digital Aquaculture Nutrition Book

Fish and Shrimp — Modern Methods, Modern Concepts

A complete visual reference for students, farm professionals and veterinary nutritionists — covering the culture cycle from broodstock egg to market harvest, and the feed chain from raw ingredient to water quality outcome.

PROF. MOUSA M. A. PREMIUM SCIENTIFIC EDITION 2026

About This Edition

The Digital Aquaculture Nutrition Book — Fish and Shrimp: Modern Methods and Modern Concepts. Premium Scientific Edition, 2026. Authored and edited by Prof. Mousa M. A., Professor of Nutrition and Clinical Nutrition, Faculty of Veterinary Medicine, Sohag University, Egypt. Part of the continuing Digital Nutrition Book series prepared for veterinary and animal-production students and for working farm professionals.

DATA INTEGRITY — THREE-CLASS LABELING

Throughout this book, every figure and table carries one of three class labels, and the label is always visible. 1 · **Published data** — values drawn from the peer-reviewed aquaculture nutrition literature and public requirement tables (NRC, FAO, species manuals). 2 · **Author-supplied farm data** — records, programs and observations from practical farm work, marked as such. 3 · **Illustrative** — worked examples and teaching constructs, **always carrying an ILLUSTRATIVE badge** so the reader never has to guess.

Requirement numbers for protein, amino acids, lipids and minerals differ between publications because they were measured on different strains, sizes, temperatures and feeding regimes. This book quotes ranges rather than single false-precision points, and it names the species each value applies to. Where evidence is thin, as with many marine finfish micronutrients, the text says so plainly instead of dressing an estimate up as a fact.

Formulation, drug use and water discharge are regulated differently in every country. The feeding programs in this book are educational templates. Adapt them to your species, your system and your local regulations, and involve a qualified aquatic animal health professional before changing any medication or health-related practice.

How the numbers are used

Diets are quoted on an as-fed or dry-matter basis as stated; energy as digestible energy (DE) unless noted; ingredient composition follows standard aquafeed tables. Rounding is intentional: a pond is not a laboratory, and the book aims for numbers you can defend in the field.



THE AUTHOR

Prof. Mousa M. A.

Professor of Nutrition and Clinical Nutrition at the Faculty of Veterinary Medicine, Sohag University, Egypt. Across a career of teaching and farm consultation, his work has followed one line: nutrition is the cheapest medicine a farm owns and the most expensive to get wrong. That conviction produced the Digital Nutrition Book series (calf, dairy cow, horse, camel, small ruminants, rabbits, plus companion volumes on nutrient metabolism), each built on the same discipline: science you can use at the pen, the tank or the pond bank.

This volume extends that discipline into water. Aquaculture is the fastest-growing animal-food sector on earth, and it is fed almost entirely from a bag or a silo: feed is the farm's largest cost and its largest source of waste. Writing it meant joining two literatures that rarely sit on the same shelf, comparative digestive physiology of fish and crustaceans and the day-to-day practice of aquafeed manufacturing and farm feeding management. Both are here, in one navigation system.

Career Focus

Clinical and applied nutrition across the species the farm actually keeps: ruminants, horses, camels, poultry, rabbits and now aquatic animals. The job is translating requirement tables into feeding decisions that hold up in the field.

From the author's desk

"A fish farmer once asked me why his feed bill had doubled while his harvest had not. The answer was not in the feed — it was in the water, the feeding table and the FCR record he never kept. This book is written so that no reader has to learn that lesson the expensive way."

Why an Aquaculture Nutrition Book?

Aquaculture now supplies more fish for human food than the world's capture fisheries, and it grows every year while wild catches stay flat. That growth came not from catching more fish but from **feeding animals in controlled systems: ponds, cages, raceways, recirculating tanks**. Feed is 50–70% of the operating cost of an intensive farm, and every kilogram of it passes through a gill, a gastric mill or a hepatopancreas before it becomes muscle, egg or waste. Nutrition is therefore not just one subject inside aquaculture; it is the discipline the whole economics hangs on.

Yet the student who leaves a general nutrition course and steps onto a fish farm meets a different world. Ruminant nutrition is built around the rumen; pig and poultry nutrition around land-locked barns. Aquaculture nutrition must answer questions those books never ask. What happens to a vitamin when it leaches from a pellet into pond water before the fish even eats it? Why does a shrimp, which moults its entire skeleton, have calcium requirements unlike any terrestrial animal? How does a recirculating system change the meaning of "feeding rate"? This book answers those questions in the order a farmer or field veterinarian actually meets them.

Modern Methods

Intensive ponds, cages and raceways; recirculating aquaculture systems (RAS); biofloc and mixed-species pond culture; automatic feeders, check trays and feeding tables.

Modern Concepts

Ideal protein and amino-acid balancing, fishmeal replacement hierarchies, functional feeds, gut microbiome management, nutrigenomics and precision feeding by sensor and model.

Two Animals, One Book

Finfish (gastric and agastric) and crustaceans (shrimp, with their moulting physiology and hepatopancreas) treated comparatively at every step, because most farms now grow both.

THE ORGANIZING IDEA

In water, feed and environment are one system. What the animal does not eat, digest or retain does not disappear; it becomes tomorrow's water-quality problem.

The chapters follow the two ribbons printed at the foot of every page. The Culture Cycle ribbon walks you through the animal's life: egg, larva, fry, juvenile, grow-out, market, broodstock. The Feed Chain ribbon walks you through the feed itself: ingredient, mill, pellet, feeding, digestion, growth, and the water that receives whatever is left. Any chapter can be read alone, but every chapter sits visibly inside the whole.

Navigation, Ribbons & Reading Paths

FIG 0.1 · THE DOUBLE RIBBON — LIVE DEMO

BELOW IS THE ACTUAL RIBBON YOU WILL SEE ON EVERY PAGE — HERE SHOWING CHAPTER 19 (SHRIMP GROW-OUT, FEED MANAGEMENT):

The **gold dot** marks the chapter's position in the Culture Cycle (top row) and the Feed Chain (bottom row). Aqua dots mark stages already covered; hollow dots mark stages ahead.

01 Follow a full production cycle

Read Parts I–II for foundations, then jump to Part IV and follow one species end to end: larval feeding (Ch 16) → grow-out (Ch 17–19) → broodstock (Ch 20).

02 Build or audit a feed

Read Part III in order: ingredients (Ch 11–12), additives (Ch 13), manufacturing (Ch 14). Then close the loop with feed management and FCR economics (Ch 15).

03 Run a modern, sustainable system

Part V is the modern-concepts core: water quality × feed (Ch 21), RAS and biofloc (Ch 22), functional feeds and immunity (Ch 23), precision feeding (Ch 24) and sustainability (Ch 25).

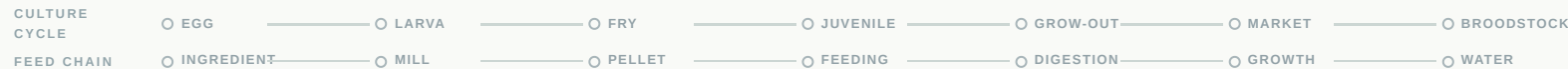
04 Solve a farm problem tonight

Go straight to the end matter: quick reference feeding tables, the troubleshooting map, the twenty golden rules and the glossary.

CHAPTER ANATOMY — EVERY CHAPTER, SAME RHYTHM

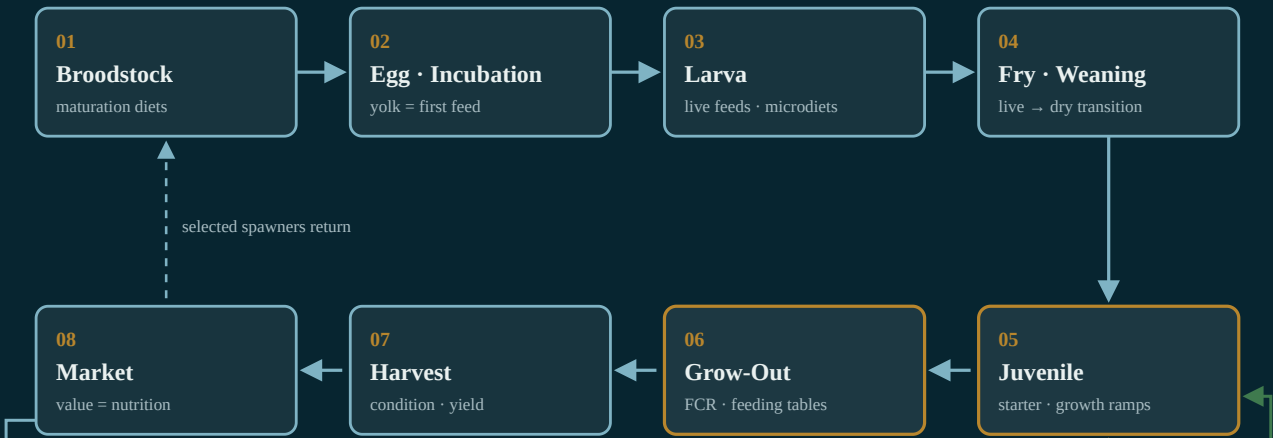
Each chapter opens with a full-bleed photograph and the key message, then delivers 3–4 dense content pages built from text, tables and figures, and closes with Field Takeaways, numbered actions you can apply at the pond bank, plus selected references where deeper reading is worth your time.

All species numbers (protein levels, temperatures, stocking densities, feeding rates) are stated for the named species and life stage; ranges reflect genuine variation across strains and environments. Constructed examples carry the **ILLUSTRATIVE** badge.



The Complete Aquaculture Nutrition Cycle

CULTURE CYCLE — THE ANIMAL'S LIFE



FEED CHAIN — INGREDIENT TO PELLET



WATER LOOP — THE SYSTEM RECEIVES EVERYTHING



HOW TO READ THE MAP

- Culture Cycle — every chapter lives at one life stage (ribbon row 1).
- Feed Chain — nutrition reaches the animal only through manufacture and delivery (ribbon row 2).
- Water Loop — what feed leaves behind returns to the farm as water quality; modern systems close this loop.

Chapters 1–25 are mapped onto these three lanes: Part I–II build the science, Part III the feed chain, Part IV the life stages, Part V the water loop and its modern contr

What This Book Holds

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Species, Systems & the Modern Core

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Student path

Read Part I → II → III in order for a complete, exam-ready foundation in fish and shrimp nutrition science.

Farm-manager path

Chapters 2, 15, 17–19, 21–22 form a working manual for running feed and water on an operating farm.

Modern-concepts path

Chapters 11–12, 22–25 cover the frontier: novel proteins, system-integrated nutrition, functional feeds, precision feeding, sustainability.